HKDSE Junior Secondary Mathematics Revision Notes
Section B: Data Handling and Statistics
香港中學文憑考試初中數學筆記
乙部：數據處理與統計學

Comprehensive Study Guide & Question Bank / 核心筆記與題庫

2026

Contents

1 Frequency Distribution Tables / 頻數分布表

Definition / 定義: Key Components of Grouped Data / 組級數據的核心要素

For data grouped into classes (e.g., $10 - 19$, $20 - 29$):

對於分組數據（例如：$10 - 19$, $20 - 29$）：

- **Class Limits (組限):** The smallest and largest data values that can belong to a class. Consists of the **Lower Class Limit** and **Upper Class Limit**.

  可歸入某組的最小及最大數據值。由**組下限**與**組上限**組成。

- **Class Boundaries (組界):** The actual numerical frontiers separating successive classes.

  $$\text{Lower Class Boundary} = \text{Lower Class Limit} - 0.5 \times \text{Accuracy Unit}$$

  $$\text{Upper Class Boundary} = \text{Upper Class Limit} + 0.5 \times \text{Accuracy Unit}$$

  分隔相鄰組$\boxed{\text{F}}$的實際數值邊界。

  $$\text{組下界} = \text{組下限} - 0.5 \times \text{准確度單位}$$

  $$\text{組上界} = \text{組上限} + 0.5 \times \text{准確度單位}$$

- **Class Mark (組中點):** The midpoint of a class interval.

  $$\begin{aligned}\text{Class Mark} &= \frac{\text{Lower Class Limit} + \text{Upper Class Limit}}{2} \\ &= \frac{\text{Lower Class Boundary} + \text{Upper Class Boundary}}{2}\end{aligned}$$

  某組區間的中央值。

  $$\text{組中點} = \frac{\text{組下限} + \text{組上限}}{2} = \frac{\text{組下界} + \text{組上界}}{2}$$

- **Class Width (組距):** The difference between the upper and lower boundaries of the same class.

  $$\text{Class Width} = \text{Upper Class Boundary} - \text{Lower Class Boundary}$$

  同一組$\boxed{\text{F}}$的組上界與組下界之差。

  $$\text{組距} = \text{組上界} - \text{組下界}$$

Worked Example / 例題: Constructing Components / 求各項統計要素

Consider the class interval $40 - 49$ for integers. Find its class limits, class boundaries, class mark, and class width.

考慮整數組$\boxed{\text{F}}$區間 $40 - 49$。求其組限、組界、組中點及組距。

Solution Space / Notes / 答題及筆記欄:

Exercise 1 (15 Problems) / 習題 1（15 題）

1. State the lower and upper class limits of the class interval 155 − 159.
   寫出組$\boxed{\text{F}}$區間 155 − 159 的組下限和組上限。

   ....

2. If data are given to 1 decimal place, find the lower class boundary of the class 10.0 − 14.9.
   若數據准確至一位小數，求組$\boxed{\text{F}}$ 10.0 − 14.9 的組下界。

   ....

3. Calculate the class mark of the interval 20 − 29.
   計算區間 20 − 29 的組中點。

   ....

4. Find the class width of the interval 35 − 44 for continuous data rounded to the nearest integer.
   對於四舍五入至最接近整數的連續數據，求區間 35 − 44 的組距。

   ....

5. A class has boundaries 14.5 and 19.5. Determine its class mark.
   某組的組界$\boxed{\text{F}}$ 14.5 及 19.5。求其組中點。

   ....

6. The class mark of an interval is 55 and its class width is 10. Find its lower and upper class limits assuming discrete integers.

   設數據🄵離散整數。已知某區間的組中點🄵 55，組距🄵 10，求其組下限和組上限。

7. Complete the missing entries in the table below:

   補全下表中的空缺項：

Class Limit / 組限	Class Boundary / 組界	Class Mark / 組中點	Class Width / 組距
$60-64$			

8. Explain why the class width of the interval $10-20$ is 11 if the data values are discrete integers, but 10 if the interval is defined via boundaries $[9.5, 19.5)$.

   若數據值🄵離散整數，解釋🄵何區間 $10-20$ 的組距是 11；但若該組是以組界 $[9.5, 19.5)$ 定義時，組距🄵是 10。

9. In a frequency table, the class marks are $12, 17, 22, 27$. What is the class width of this distribution?

   在某頻數表中，組中點分🄵🄵 $12, 17, 22, 27$。求此分🄵的組距。

10. If the upper class boundary of a class is 49.5 and the lower class boundary is 39.5, write down its corresponding integer class limits.

    若某組的組上界□ 49.5，組下界□ 39.5，寫出其對應的整數組限。

11. **(DSE Level)** A data set contains weights measured to the nearest 0.1 kg. If one class is 50.0 – 54.9 kg, state its exact class boundaries.

    (文憑試程度) 一組數據包含准確至最接近 0.1 kg 的重量。若其中一組是 50.0 – 54.9 kg，指出其准確的組界。

12. Form a frequency table with 4 classes of equal width for the following data: 12, 24, 15, 31, 28, 19, 33, 38. Use 10 – 19 as the first class.

    □以下數據建立一個含有 4 個等距組□的頻數表：12, 24, 15, 31, 28, 19, 33, 38。以 10 – 19 □第一組。

13. A distribution has class marks $105, 115, 125$. Find the boundaries for the second class.

    某分□的組中點□ $105, 115, 125$。求第二組的組界。

14. True or False: "The difference between successive class marks equals the class width." Justify your answer.

    是非題:「相鄰組中點的差必等於組距。」試正其🄵。

    ....

15. Given that the lower class boundary of a class is L, and its class width is W, express its class mark in terms of L and W.

    已知某組的組下界🄵 L, 組距🄵 W, 以 L 和 W 表達該組的組中點。

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2 Histograms, Frequency Polygons, and Curves / 直方圖、頻數多邊形及頻數曲

Definition / 定義: Histograms vs. Frequency Polygons / 直方圖與頻數多邊形

- **Histogram (直方圖):** A graphical representation of grouped continuous data where **Class Boundaries** are marked on the horizontal axis and **Frequency** on the vertical axis. The bars must touch each other without gaps.

  一種用於表示分組連續數據的統計圖。其中軸代表**組界**，縱軸代表**頻數**。各長方形條形必須緊貼，中間不留空隙。

- **Frequency Polygon (頻數多邊形):** A line graph formed by connecting the midpoints of the tops of the bars in a histogram. It must start and end on the horizontal axis at the class marks of two imaginary classes with zero frequency.

  將直方圖中各條形頂部的中點連接而成的折圖。它必須在軸上以兩個頻數零的擬組的組中點作起點和終點。

Worked Example / 例題: Interpreting a Histogram / 解讀直方圖

Sketch an outline of a histogram for two classes: 10 – 14 (freq: 5) and 15 – 19 (freq: 8).
試草擬包含以下兩組的直方圖輪廓：10 – 14（頻數：5）及 15 – 19（頻數：8）。

Solution Space / Notes / 答題及筆記欄:

Exercise 2 (15 Problems) / 習題 2（15 題）

1. On which axis of a histogram do we plot the class boundaries?
   在直方圖中，組界應繪畫在全幅圖的哪一條軸上?

   ....

2. What values are represented on the horizontal axis when plotting points to form a frequency polygon?
   在描點繪$\boxed{\text{F}}$頻數多邊形時，$\boxed{\text{F}}$軸代表甚$\boxed{\text{F}}$數值?

   ....

3. If a histogram has bars with class boundaries $10.5 - 15.5$ and $15.5 - 20.5$, what is the class width?
   若某直方圖條形的組界$\boxed{\text{F}}$ $10.5 - 15.5$ 與 $15.5 - 20.5$，求組距。

   ....

4. Explain why there are no gaps between the bars of a histogram.
   解釋$\boxed{\text{F}}$何直方圖的條形之間$\boxed{\text{F}}$有空隙。

   ....

5. A frequency polygon contains the point $(24.5, 12)$. What is the class mark and frequency of this class?
   某頻數多邊形包含點 $(24.5, 12)$。求該組的組中點及頻數。

   ....

6. To close a frequency polygon on the horizontal axis at the lower end when the first class mark is 15 and class width is 10, what point must be plotted?

   若第一組的組中點$\boxed{F}$ 15，組距$\boxed{F}$ 10，$\boxed{F}$使頻數多邊形在左端（下端）延伸至$\boxed{F}$軸閉合，應描哪一點?

   . . . .

7. A histogram has equal class widths of 5. If the height of the bar for class $20 - 24$ is 12 units, what is its frequency?

   某直方圖的等距組距$\boxed{F}$ 5。若組$\boxed{F}$ $20 - 24$ 的條形高度$\boxed{F}$ 12 單位，求其頻數。

   . . . .

8. If the total area under a frequency polygon represents the total frequency $\times$ class width, calculate the total area if $N = 40$ and class width $= 6$.

   若頻數多邊形下方的總面積代表「總頻數 $\times$ 組距」，且總頻數 $N = 40$，組距 $= 6$，計算該總面積。

   . . . .

9. Distinguish between a frequency polygon and a frequency curve.

   試分辨頻數多邊形與頻數曲$\boxed{F}$的分$\boxed{F}$。

   . . . .

10. Convert the following frequency polygon points into class boundaries: $(12, 4), (17, 9), (22, 5)$.

將以下頻數多邊形的點轉□□各組的組界：$(12, 4), (17, 9), (22, 5)$。

....

11. In a histogram, if the frequency of a class is doubled while its width remains unchanged, how does the bar change?

在直方圖中，若某組的頻數增加一倍而組距不變，其條形會如何改變？

....

12. **(DSE Level)** A histogram has an error: the classes are $10 - 19$ and $20 - 39$. If both have a frequency of 10, explain why making their heights equal is incorrect.

(文憑試程度) 某直方圖出錯：組□分□□ $10 - 19$ 和 $20 - 39$。若兩組的頻數皆□ 10，解釋□何將兩者的條形高度設□相同是不正確的。

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13. Find the coordinates of the vertices of the frequency polygon for the data: Class $1 - 5$ (freq 3), Class $6 - 10$ (freq 7).

求以下數據的頻數多邊形頂點（頂點坐標）：組□ $1 - 5$（頻數 3），組□ $6 - 10$（頻數 7）。（需包括與□軸相交的端點）

....

14. Why must the total area under a histogram equal the total area under its corresponding frequency polygon?

$\boxed{\text{F}}$何直方圖下的總面積必定等於其對應頻數多邊形下的總面積?

....

15. Sketch a basic frequency polygon given the class boundaries $0.5-5.5$ (freq 4), $5.5-10.5$ (freq 8), and $10.5-15.5$ (freq 2). Include the anchor points.

根據以下數據草擬一頻數多邊形：組界 $0.5-5.5$（頻數 4）、$5.5-10.5$（頻數 8）及 $10.5-15.5$（頻數 2）。必須標記與$\boxed{\text{F}}$軸相交的端點。

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3 Cumulative Frequency Polygons and Quartiles / 累積頻數多邊形及四分位數

Definition / 定義: Cumulative Frequency and Quantiles / 累積頻數與分位數

- **Cumulative Frequency Polygon / Ogive (累積頻數多邊形):** A line graph where the horizontal axis represents the **Upper Class Boundaries** and the vertical axis represents the **Cumulative Frequency**. It always starts at (Lower Boundary of first class, 0).

  一種折□圖，其中□軸代表**組上界**，縱軸代表**累積頻數**。起點必定□（第一組的組下界，0）。

- **Median (Q_2, 中位數):** The 50th percentile value read from the graph.

  從統計圖上讀取的第 50 百分位數值。

- **Lower Quartile (Q_1, 下四分位數):** The 25th percentile value.

  第 25 百分位數值。

- **Upper Quartile (Q_3, 上四分位數):** The 75th percentile value.

  第 75 百分位數值。

- **Percentile (P_k, 百分位數):** The value below which $k\%$ of the data falls.

  有 $k\%$ 的數據低於或等於該值的特定數值。

Worked Example / 例題: Reading an Ogive / 解讀累積頻數多邊形

Given a cumulative frequency curve with total frequency $N = 80$. State the cumulative frequencies used to locate Q_1, Q_2, and Q_3.
已知一累積頻數曲□的總頻數为 $N = 80$。指出用作找出 Q_1、Q_2 及 Q_3 所對應的累積頻數值。

Solution Space / Notes / 答題及筆記欄:

Exercise 3 (15 Problems) / 習題 3（15 題）

1. Which values are plotted on the horizontal axis of a cumulative frequency polygon?
   累積頻數多邊形的$\boxed{F}$軸繪畫的是甚$\boxed{F}$數值?

2. If the total frequency of a data set is 120, what is the cumulative frequency corresponding to the median?
   若一組數據的總頻數$\boxed{F}$ 120，中位數對應的累積頻數是多少?

3. Find the cumulative frequency rank required to determine the 35th percentile of 200 data items.
   求確定 200 個數據項中第 35 百分位數所需的累積頻數位置。

4. A cumulative frequency polygon starts at the point $(14.5, 0)$. What is the lower boundary of the first class?
   某累積頻數多邊形始於點 $(14.5, 0)$。求第一組的組下界。

5. From an ogive, $Q_1 = 42$ and $Q_3 = 68$. Calculate the Interquartile Range (IQR).
   從累積頻數多邊形中讀得 $Q_1 = 42$ 及 $Q_3 = 68$。計算四分位數間距（IQR）。

6. If the total number of students is 60, and 45 students scored below 70 marks, what percentile rank corresponds to a score of 70?

   若學生總數🄵 60 人，且有 45 人的分數低於 70 分，分數 70 對應的是哪一個百分位數?

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7. Why does a cumulative frequency polygon never trend downwards?

   🄵何累積頻數多邊形的圖🄵🄵不會向下傾斜?

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8. Given the table below, find the cumulative frequencies for each class:

   根據下表，求各組的累積頻數:

Class Limits / 組限	Frequency / 頻數	Cumulative Frequency / 累積頻數
10 – 19	4	
20 – 29	11	
30 – 39	5	

   ....

9. Based on Question 8, state the coordinates of the four points used to plot the cumulative frequency polygon.

   根據第 8 題，指出繪🄵該累積頻數多邊形所需的四個點的坐標。

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10. From an ogive, the 80th percentile is found to be 56.5 kg. Interpret this statement in words.

從累積頻數曲🄵中，測得第 80 百分位數🄵 56.5 kg。用文字解釋這句陳述的統計意義。

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11. **(DSE Level)** In a test, the passing mark is set such that the top 30% of students pass. If $N = 150$, what cumulative frequency value should you look up on the vertical axis to find the passing mark?

(文憑試程度) 某測驗的合格分數設🄵令前 30% 的學生合格。若總人數 $N = 150$，你應該在縱軸上查找哪一個累積頻數值以求得合格分數?

....

12. If $Q_1 = 20$, $Q_2 = 35$, and $Q_3 = 40$, describe the skewness of the data distribution.

若 $Q_1 = 20$, $Q_2 = 35$, 及 $Q_3 = 40$，試描述該數據分🄵的偏度（偏斜狀態）。

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13. An ogive passes through $(20.5, 10)$ and $(30.5, 30)$. Estimate the number of data items falling within the interval $20.5 - 30.5$.

某累積頻數多邊形通過 $(20.5, 10)$ 與 $(30.5, 30)$。估計落在區間 $20.5 - 30.5$ 🄵的數據項數目。

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14. Find the percentile rank of a value if it lies exactly at the upper quartile.
若某一數值恰好等於上四分位數，求其百分位數位置。

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15. Sketch a general shape of a cumulative frequency curve for a normal distribution, labeling axes.
試草擬正態分🄵數據的累積頻數曲🄵的一般形狀，🄵標記兩軸。

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4 Organisation of Data and Diagrams / 數據的組織與統計圖

Definition / 定義: Statistical Diagrams / 常用統計圖

- **Stem-and-Leaf Diagram (干葉圖):** A method to split each data value into a ”Stem” (leading digits) and a ”Leaf” (trailing digit) to show the distribution shape while preserving raw data.

  將每個數據值拆分□「干」(前導數字)與「葉」(尾隨個位數字)的方法。它既能顯示分□形狀，又能保留原始數據。

- **Back-to-Back Stem-and-Leaf Diagram (背靠背干葉圖):** Used to compare two data sets using the same stems, with leaves spreading outwards left and right.

  用於比較共享同一組「干」的兩組數據，其「葉」分□向左和向右兩側擴展。

- **Pie Chart, Bar Chart, Broken-line Graph (圓形圖、條形圖、折□圖):** Basic charts representing proportions, categories, and trends over time respectively.

  分□用於表示各部分所□比例、類□頻數比較，以及數據隨時間變化的趨勢。

Worked Example / 例題: Reading Stem-and-Leaf / 解讀干葉圖

Given the stem-and-leaf plot:
已知下列干葉圖：

Stem (tens) / 干（十位）	Leaf (units) / 葉（個位）
2	1, 3, 5
3	0, 4

List all the data values and find the total number of items.
列出所有原始數據值□求數據總項數。

Solution Space / Notes / 答題及筆記欄:

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Exercise 4 (15 Problems) / 習題 4（15 題）

1. For the data value 145, if the stem represents hundreds and tens, write down the leaf.
   對於數據值 145，若「干」代表百位及十位，寫出其「葉」的值。

2. List the raw data from this row of a stem-and-leaf plot: `4 | 0 2 2 7` (Key: `4 | 0` means 40).
   列出干葉圖中這一行的所有原始數據：`4 | 0 2 2 7`（附$\boxed{\text{F}}$：`4 | 0` 代表 40）。

3. State one advantage of a stem-and-leaf diagram over a histogram.
   寫出干葉圖比直方圖更具優勢的一個特點。

4. Find the median of the following data directly from the stem-and-leaf plot:
   直接從以下干葉圖中找出數據的中位數：

Stem / 干	Leaf / 葉
0	5, 8
1	2, 3, 7
2	1, 6

5. In a pie chart, a sector representing 15 out of 60 students has what central angle?
在圓形圖中，代表 60 人中□ 15 人的扇形，其圓心角是多少?

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6. A sector in a pie chart has a central angle of 72°. What percentage of the total data does this sector represent?
圓形圖中某扇形的圓心角□ 72°。該扇形代表總數據的百分之幾?

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7. Explain when a broken-line graph is more appropriate than a bar chart.
解釋在甚□情□下使用折□圖會比條形圖更合適。

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8. Construct a stem-and-leaf diagram for: 23, 45, 28, 31, 39, 20, 42, 33.
□以下數據繪□一個干葉圖：23, 45, 28, 31, 39, 20, 42, 33。

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9. For a back-to-back stem-and-leaf diagram, Group A's row is `3 1 | 5 | 2 4`. Write down the values for both groups belonging to stem 5.

在一背靠背干葉圖中，某行記□□ A 組 `3 1 | 5 | 2 4` B 組。寫出兩組中屬於「干」□ 5 的所有原始數據值。

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10. Find the range of Class B from the following back-to-back diagram:

從以下背靠背干葉圖中，求 B 班數據的極差（全距）：

Class A / A 班	Stem / 干	Class B / B 班
9, 2	1	3, 5, 8
4, 0	2	1, 7

....

11. **(DSE Level)** A back-to-back stem-and-leaf diagram compares the test marks of two classes. If Class X has 25 students and Class Y has 25 students, how do you locate the median for each class?

(文憑試程度) 一幅背靠背干葉圖用於比較兩班的測驗分數。若 X 班有 25 人，Y 班亦有 25 人，你應該如何分□定位兩班的中位數？

....

12. If a leaf in a stem-and-leaf diagram has 0 items in a particular stem row, how should that row be presented?

    在干葉圖中，若某一「干」行對應□有任何「葉」數據項，該行應如何展示？

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13. A bar chart shows category frequencies. If the scale on the vertical axis does not start at 0, what misinterpretation can occur?

    某條形圖展示了各類□的頻數。若縱軸的刻度不是由 0 開始，可能會引起甚□誤導或誤解？

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14. Convert these percentages into sector angles for a pie chart: Category A (40%), Category B (35%), Category C (25%).

    將以下百分比轉□□圓形圖的扇形圓心角：類□ A (40%)、類□ B (35%)、類□ C (25%)。

    ....

15. Given the raw data: 102, 114, 105, 121, 118, 120, 109. Draw a suitable stem-and-leaf diagram.

    已知原始數據：102, 114, 105, 121, 118, 120, 109。試繪□一合適的干葉圖。

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5 Measures of Central Tendency and Dispersion / 集中趨勢及離散程度的量度

Definition / 定義: Central Tendency vs Dispersion / 集中趨勢與離散程度

- **Mean ($\bar{x}$, 平均值):** $\bar{x} = \frac{\sum x_i}{n}$. For grouped data, $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, where x_i is the class mark.
  $\bar{x} = \frac{\sum x_i}{n}$。分組數據則F $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$，其中 x_i 代表組中點。
- **Mode / Modal Class (F數 / F數組):** The value or class interval with the highest frequency.
  出現頻數最高（最多）的數據值或組F區間。
- **Median (中位數):** The middle value when sorted.
  數據按大小順序排列後，位處中央的數值。
- **Range (極差 / 全距):** Maximum Value − Minimum Value.
  最大值 − 最小值。
- **Interquartile Range (IQR, 四分位數間距):** $Q_3 - Q_1$.
  $Q_3 - Q_1$（上四分位數與下四分位數之差）。
- **Variance (σ^2, 方差):** $\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum x_i^2}{n} - \bar{x}^2$.
  衡量數據分散程度的量度。
- **Standard Deviation (σ, 標准差):** $\sigma = \sqrt{\text{Variance}}$.
  $\sigma = \sqrt{\text{方差}}$。

Property or Theorem / 性質或定理: Effects of Data Transformation / 數據變F的影響

If a constant k is added to every data value:

$$\text{New Mean} = \bar{x} + k, \quad \text{New Median} = \text{Median} + k, \quad \text{New SD} = \sigma$$

If every data value is multiplied by a positive constant c:

$$\text{New Mean} = c\bar{x}, \quad \text{New Median} = c \times \text{Median}, \quad \text{New SD} = c\sigma$$

若每個數據值均加上常數 k:

$$\text{新平均值} = \bar{x} + k, \quad \text{新中位數} = \text{中位數} + k, \quad \text{新標准差} = \sigma$$

若每個數據值均乘以正常數 c:

$$\text{新平均值} = c\bar{x}, \quad \text{新中位數} = c \times \text{中位數}, \quad \text{新標准差} = c\sigma$$

Worked Example / 例題: Calculating Summary Statistics / 計算統計量值

Find the mean, mode, range, and standard deviation of the data set: $2, 4, 4, 8, 12$.
求數據集 $2, 4, 4, 8, 12$ 的平均值、F數、極差及標准差。

Solution Space / Notes / 答題及筆記欄:

. . . .

Exercise 5 (15 Problems) / 習題 5（15 題）

1. Find the mean of 5, 8, 12, 15, and 20.
   求 5, 8, 12, 15, 20 的平均值。

   . . . .

2. Determine the mode of the data: 3, 4, 4, 5, 6, 6, 6, 7.
   求數據 3, 4, 4, 5, 6, 6, 6, 7 的F數。

   . . . .

3. Arrange the data and find the median: 14, 9, 18, 12, 15.
   排列數據F找出中位數：14, 9, 18, 12, 15。

   . . . .

4. Calculate the range of the set: $-3, 2, 8, 1, 15$.
   計算數據集 $-3, 2, 8, 1, 15$ 的極差。

5. A data set has $Q_1 = 15$ and $Q_3 = 37$. Find the interquartile range.
   某數據集的 $Q_1 = 15$ 且 $Q_3 = 37$。求四分位數間距。

6. If the variance of a distribution is 49, what is its standard deviation?
   若某分$\boxed{\text{F}}$的方差$\boxed{\text{F}}$ 49，其標准差是多少?

7. A set of numbers has a mean of 50. If 4 is added to every number, find the new mean.
   一組數字的平均值$\boxed{\text{F}}$ 50。若每個數字都加上 4，求新的平均值。

8. The standard deviation of a set of scores is 6. If each score is multiplied by 3, find the new standard deviation.
   某組分數的標准差$\boxed{\text{F}}$ 6。若每個分數都乘以 3，求新的標准差。

9. Find the modal class for the distribution: $10 - 14$ (freq 5), $15 - 19$ (freq 12), $20 - 24$ (freq 8).
求此分$\boxed{\text{F}}$的$\boxed{\text{F}}$數組：$10 - 14$（頻數 5）、$15 - 19$（頻數 12）、$20 - 24$（頻數 8）。

....

10. Calculate the estimated mean for the grouped data in Question 9 using class marks.
利用組中點，計算第 9 題中分組數據的估計平均值。

....

11. Find the standard deviation of the values: 10, 10, 10, 10. Explain your answer.
求數值 10, 10, 10, 10 的標准差，$\boxed{\text{F}}$解釋你的答案。

....

12. **(DSE Level)** If the variance of $x_1, x_2, \ldots, x_n$ is V, write down the variance of $2x_1 - 3, 2x_2 - 3, \ldots, 2x_n - 3$ in terms of V.
(文憑試程度) 若 $x_1, x_2, \ldots, x_n$ 的方差是 V，試以 V 寫出 $2x_1 - 3, 2x_2 - 3, \ldots, 2x_n - 3$ 的方差。

....

13. Which measure of central tendency is most affected by extreme outliers? Why?
哪一個集中趨勢的量度最易受極端值（離群值）影響？F甚F?

....

14. For the numbers 2, 3, 5, 7, 11, find the upper quartile (Q_3).
求數字 2, 3, 5, 7, 11 的上四分位數（Q_3）。

....

15. Given $\sum_{i=1}^{5} x_i = 50$ and $\sum_{i=1}^{5} x_i^2 = 520$, calculate the standard deviation of these 5 numbers.
已知 $\sum_{i=1}^{5} x_i = 50$ 且 $\sum_{i=1}^{5} x_i^2 = 520$，計算這 5 個數字的標準差。

....

6 Finding Unknown Data from Given Measures / 從已知量值求未知數據

Definition / 定義: Solving for Unknowns / 反求未知數的方法

By working backward from statistical definitions, missing individual values or frequencies can be determined using algebraic setups:

$$\sum x_i = n \times \bar{x}$$
$$\text{Missing Value} = (\text{New } n \times \text{New Mean}) - (\text{Old } n \times \text{Old Mean})$$

通過從統計學定義進行逆向運算，可以利用代數方程建立來求得缺失的個🄵數據值或頻數：

$$\sum x_i = n \times \bar{x}$$
$$\text{缺失的數據值} = (\text{新項數}n \times \text{新平均值}) - (\text{舊項數}n \times \text{舊平均值})$$

Worked Example / 例題: Finding a Missing Value / 求未知項

The mean of four numbers $6, 8, 12, x$ is 10. Find the value of x.
四個數字 $6, 8, 12, x$ 的平均值🄵 10。求 x 的值。

Solution Space / Notes / 答題及筆記欄:

. . . .

Exercise 6 (15 Problems) / 習題 6（15 題）

1. The mean of three numbers is 20. Two of the numbers are 15 and 22. Find the third number.

   三個數字的平均值是 20。其中兩個數字是 15 和 22。求第三個數字。

2. A set of numbers $3, 5, 7, 8, k$ is arranged in ascending order. If the median is 7, what are the possible integer values of k?

   一組數字 $3, 5, 7, 8, k$ 已按遞增順序排列。若中位數是 7，求 k 所有可能的整數值。

3. The mode of the data set $2, 5, 5, 8, 8, x$ is 8. Find x.

   數據集 $2, 5, 5, 8, 8, x$ 的□數是 8。求 x。

4. The range of the set $\{12, 7, 4, 19, x\}$ is 17. If x is the maximum value, find x.

   數據集 $\{12, 7, 4, 19, x\}$ 的極差是 17。若 x 是最大值，求 x。

5. The mean of 9 items is 40. When an 10th item is added, the new mean becomes 42. Find the value of the 10th item.

   9 項數據的平均值□ 40。當加入第 10 項數據後，新平均值變□ 42。求第 10 項數據的值。

6. Six numbers have a mean of 15. If a number is removed, the remaining five numbers have a mean of 13. What number was removed?

   六個數字的平均值$\boxed{\text{F}}$ 15。若移去一個數字，剩下的五個數字的平均值變$\boxed{\text{F}}$ 13。問移去了哪一個數字？

   ....

7. The data set $4, 7, x, 12, 15$ is sorted in ascending order. If the mean equals the median, find x.

   數據集 $4, 7, x, 12, 15$ 已按遞增順序排列。若平均值等於中位數，求 x。

   ....

8. A data set consists of $5, 8, 12, y$. If the interquartile range is 5 and $y > 12$, determine y.

   某數據集包含 $5, 8, 12, y$。若其四分位數間距$\boxed{\text{F}}$ 5 且 $y > 12$，求 y 的值。

   ....

9. The mean of a, b, c is 10. The mean of a, b, c, d, e is 12. Find the mean of d and e.

   a, b, c 的平均值是 10。a, b, c, d, e 的平均值是 12。求 d 和 e 的平均值。

   ....

10. Four distinct positive integers have a median of 6.5 and a mode does not exist. If the smallest number is 2, what is the largest possible value of the maximum number if the mean is 7?

    四個互不相同的正整數，其中位數🄵 6.5 且不存在🄵數。若最小的數字是 2，且平均值是 7，問最大數字的最大可能值是多少?

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11. **(DSE Level)** A frequency table has two missing values a and b:

Value / 數值	1	2	3
Frequency / 頻數	4	a	b

    If the total frequency is 10 and the mean is 2.1, find a and b.

    (文憑試程度) 某頻數表有兩個缺失值 a 和 b。若總頻數🄵 10 且平均值🄵 2.1，求 a 和 b 的值。

    ....

12. The numbers x, y, z have a mean of 8 and a variance of 0. Find the values of x, y, and z.

    數字 x, y, z 的平均值🄵 8 且方差🄵 0。求 x, y, z 的值。

    ....

13. The range of three positive integers is 8. Their sum is 15. Find the three numbers if their mode is unique.

三個正整數的極差□ 8，它們的和是 15。若它們有唯一的□數，求這三個數字。

....

14. If the mean of x_1, x_2, x_3, x_4 is $\bar{x}$, and $\sum_{i=1}^{3} x_i = S$, express x_4 in terms of $\bar{x}$ and S.

若 x_1, x_2, x_3, x_4 的平均值是 $\bar{x}$，且 $\sum_{i=1}^{3} x_i = S$，以 $\bar{x}$ 和 S 表達 x_4。

....

15. A set of 5 data entries has a mean of 6 and a standard deviation of 2. If a new data entry equal to 6 is added to the set, find the new standard deviation.

一組含有 5 個數據的資料，其平均值□ 6，標準差□ 2。若加入一個數值同樣□ 6 的新數據，求新的標準差。

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